Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI Agentic Systems · Context Engineering · LLM Evaluation · LangGraph · Python · Production AI · RAG Pipelines

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SUMMARY

Product-minded Senior Software Engineer with 9+ years of production engineering depth and two years building agentic AI systems across all four pillars of Nordstrom's AI Agentic Solutions team. Built **Leasel**: 8-agent LangGraph system with tool-use pipelines, multi-agent orchestration, context engineering (token-aware retrieval assembly, hybrid search, citation grounding), memory and state management (durable workflow state, session context, SQLite knowledge repository), and HITL escalation - responsible deployment enforced in code throughout. Built **AgentProbe**: open-source LLM evaluation toolkit with offline benchmarks, online telemetry via CI regression harness, LLM-as-judge scoring, and failure-mode analysis - keeps production AI quality from degrading after deployment. At Entrata (9+ years, 100K+ monthly active users), owned mission-critical production systems end-to-end across the full SDLC in a regulated environment. Strong Python, LangGraph, RAG pipelines, AWS, CI/CD, and AI-assisted development. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agentic AI · Multi-Agent Orchestration | LangGraph · LangChain | Context Engineering · RAG Pipelines | Memory & State Management | LLM Evaluation · Offline Benchmarks | Guardrails · Responsible Deployment | Tool-Use · HITL Workflows | Python · FastAPI · Django | Prompt Engineering · Structured Outputs | Embedding Models · Vector Stores | Anthropic · OpenAI · Azure OpenAI | AWS (S3 · EC2 · RDS · Lambda) | Docker · Kubernetes · CI/CD | Observability · Failure-Mode Analysis | PostgreSQL · Redis · SQLite | Open-Source AI Contributions | Distributed Systems | Full-Stack · REST APIs

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, AI-powered features, and data pipelines end-to-end; surfacing insights from raw business data into application-facing APIs and reporting layers.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Built a production agentic workflow (LangGraph StateGraph) with context engineering over regulatory documents, tool-use pipelines, validation guardrails, and conditional routing - natural-language inputs to structured outputs; deployed MCP server connecting agent outputs to downstream enterprise systems with defined access boundaries and observability.
- Engineered full document ingestion, chunking, retrieval, and FastAPI streaming pipeline with responsible deployment practices and retry semantics.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, multi-tenant)

- **Owned production systems end-to-end across the full SDLC** - requirements, design, implementation, testing, deployment, and production operations for 7+ years serving 100K+ users in a regulated environment; zero compliance violations, zero security incidents.
- **Led production incident response and failure-mode analysis** - built internal debugging tools reducing debugging time by 40%; led postmortems and root-cause analysis across distributed services; drove durable fixes and resiliency improvements; maintained bug count below 10 for 6+ consecutive months.
- **Drove product-minded engineering excellence** through 300+ peer code reviews, design reviews, and automated test pipelines; Python/Bash automation eliminated 80% of manual operational workload and accelerated team delivery velocity.
- Owned HUD compliance data pipeline (few thousand federal rows → ~180,000 domain records), RabbitMQ-based async distributed processing, PHI-encrypted AWS S3 migration (100% coverage), and 200+ SQL query optimizations (30% gain) across complex multi-tenant PostgreSQL schemas.
- Led microservices redesign (Docker) - 50% scalability improvement, 30% deployment time reduction; mentored 5+ engineers; shipped 20+ full-stack modules across 120+ Agile sprints.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built property management modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration from legacy storage; delivered large-scale data import workflows (1.6M records).

PROJECTS

Leasel - Production Agentic AI System (All Four Pillars)

2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Agent Engineering:** 8-agent LangGraph StateGraph with durable workflow state, tool-use pipelines, multi-agent orchestration (Triage → Retrieval → Configuration → Historical Incident → Reasoning → Validation → Human Escalation), and HITL escalation when confidence is insufficient. **Context Engineering:** token-aware RAG assembly - OCR-fallback ingestion, hybrid search (TF-IDF + semantic reranking), citation grounding to exact source clauses, missing/conflicting clause detection. **Memory & State Management:** durable workflow state, short-term session context, long-term SQLite knowledge repository (documents, settings, incidents, release notes, audit logs). **Responsible Deployment:** hard safety boundaries enforced in code - LLM reasons, deterministic or human logic makes every regulated call; audit logging on every agent decision.

- 93.6% precision and 87.9% F1 across 1,240 test cases (10 compliance rules, 5 US states + federal).

AgentProbe - Open-Source LLM Evaluation & Guardrails Toolkit 2025 I Python, FastAPI, LangGraph, CI/CD I github.com/venkatalakshmiparimala/AgentProbe

- **Offline benchmarks + online telemetry** - CI-integrated regression harness with LLM-as-judge scoring and golden datasets catches agent degradations before they reach production; 15-endpoint FastAPI backend; 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); multi-provider LLM routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama); failure-mode analysis built into every evaluation run.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017